

Experiment 2

Design and Verification of 8×1 Multiplexer and 8×3 Encoder using Basic Logic Gates on Logisim

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1. Aim / Objective

To design and verify the operation of an **8×1 Multiplexer (MUX)** and an **8×3 Encoder** using basic logic gates on the Logisim simulation software, and to validate the outputs against the expected truth tables.

2. Theory

2.1 Multiplexer (8×1 MUX)

A multiplexer is a combinational circuit that selects one of many input lines and forwards it to a single output line based on select signals. An **8×1 MUX** has 8 data inputs (I_0 – I_7), 3 select lines (S_2 , S_1 , S_0), and 1 output (Y). The select lines determine which input is routed to the output.

Boolean Expression: $Y = S_2 \cdot S_1 \cdot S_0 \cdot I_0 + S_2 \cdot S_1 \cdot S_0 \cdot I_1 + \dots + S_2 \cdot S_1 \cdot S_0 \cdot I_7$

2.2 Encoder (8×3 / Octal-to-Binary Encoder)

An encoder is a combinational circuit that converts 2^n input lines into n output lines. An **8×3 encoder** (octal-to-binary) takes 8 input lines (I_0 – I_7 , with only one active HIGH at a time) and produces a 3-bit binary code (A_2 , A_1 , A_0) at the output.

Boolean Expressions: $A_2 = I_4 + I_5 + I_6 + I_7$ $A_1 = I_2 + I_3 + I_6 + I_7$ $A_0 = I_1 + I_3 + I_5 + I_7$

3. Truth Tables

3.1 8×1 Multiplexer Truth Table

S ₂	S ₁	S ₀	Output (Y)
0	0	0	I ₀
0	0	1	I ₁
0	1	0	I ₂
0	1	1	I ₃
1	0	0	I ₄
1	0	1	I ₅
1	1	0	I ₆
1	1	1	I ₇

3.2 8×3 Encoder Truth Table

I7	I6	I5	I4	I3	I2	I1	I0	A2	A1	A0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

4. Components / Gates Required

Component	8×1 MUX	8×3 Encoder
NOT Gates	3	—
AND Gates	8 (3-input)	—
OR Gates	1 (8-input)	3 (4-input)
Total Gates	12	3

5. Procedure

1. Open Logisim and create a new project file.
2. For the 8×1 MUX: Place 3 input pins as select lines (S2, S1, S0) and add NOT gates to generate their complements.
3. Add 8 three-input AND gates, each connected to the appropriate combination of select lines and a data input (I0–I7).
4. Connect all AND gate outputs to an 8-input OR gate to produce the final output Y.
5. Verify the MUX output by toggling select lines and checking which input is forwarded.
6. For the 8×3 Encoder: Place 8 input pins (I0–I7, only one HIGH at a time).
7. Connect inputs to three OR gates to generate outputs A2, A1, and A0 as per the Boolean expressions.
8. Also implement the Priority Encoder variant using Logisim's built-in priority encoder component.
9. Verify all outputs against the truth tables for both circuits.
10. Take screenshots of each completed and verified circuit.

6. Logisim Circuit Screenshots

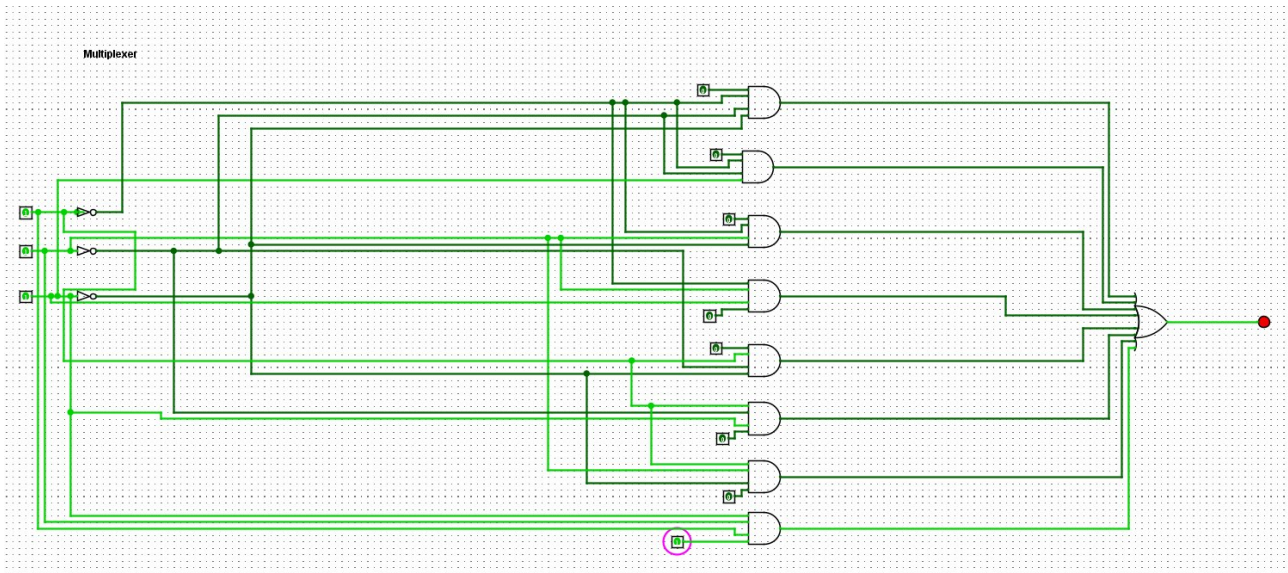


Fig. 1 – 8×1 Multiplexer implemented using basic logic gates (AND, NOT, OR) on Logisim.

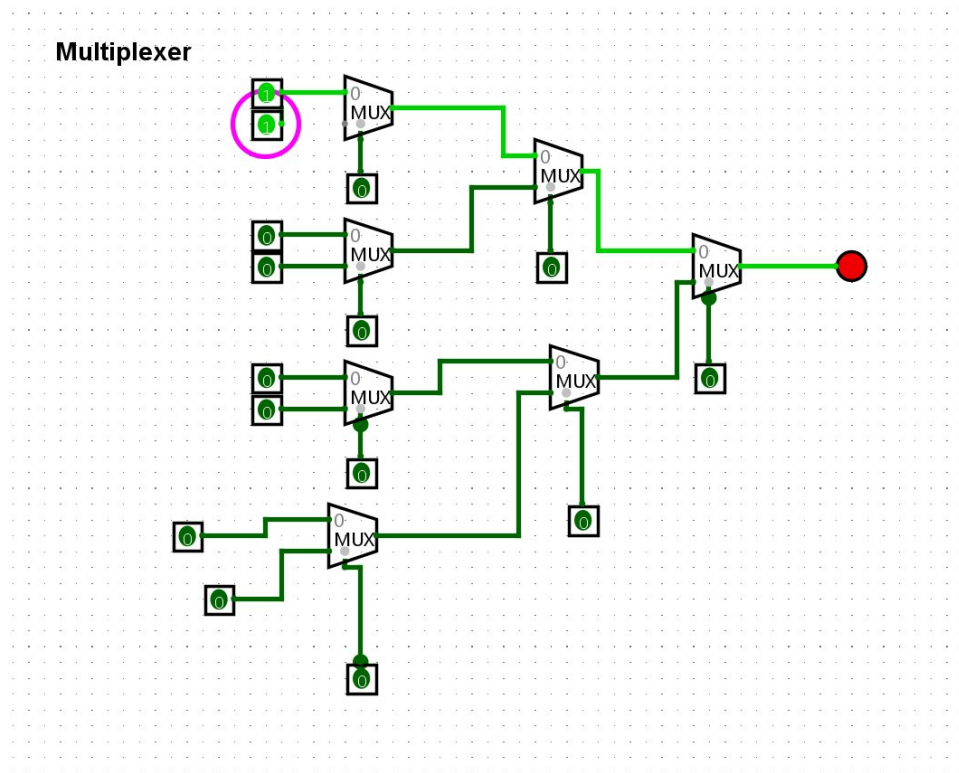


Fig. 2 – 8×1 Multiplexer implemented using cascaded MUX blocks on Logisim.

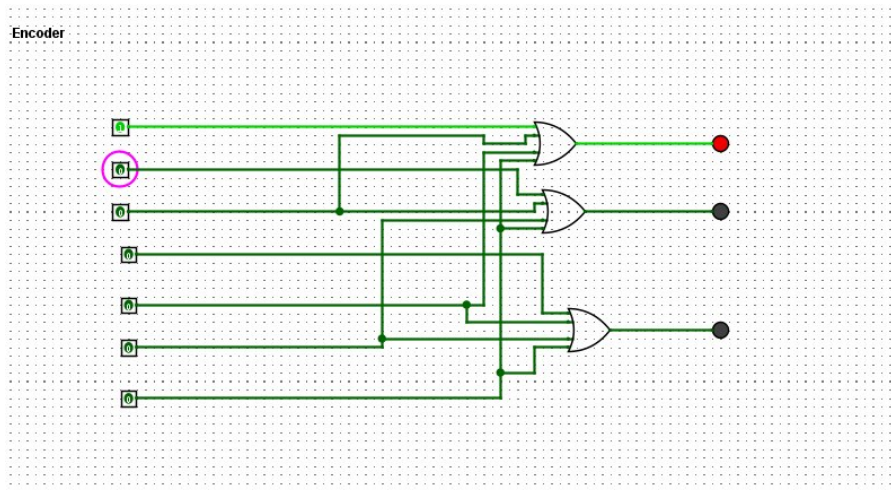


Fig. 3 – 8×3 Encoder implemented using OR gates on Logisim.

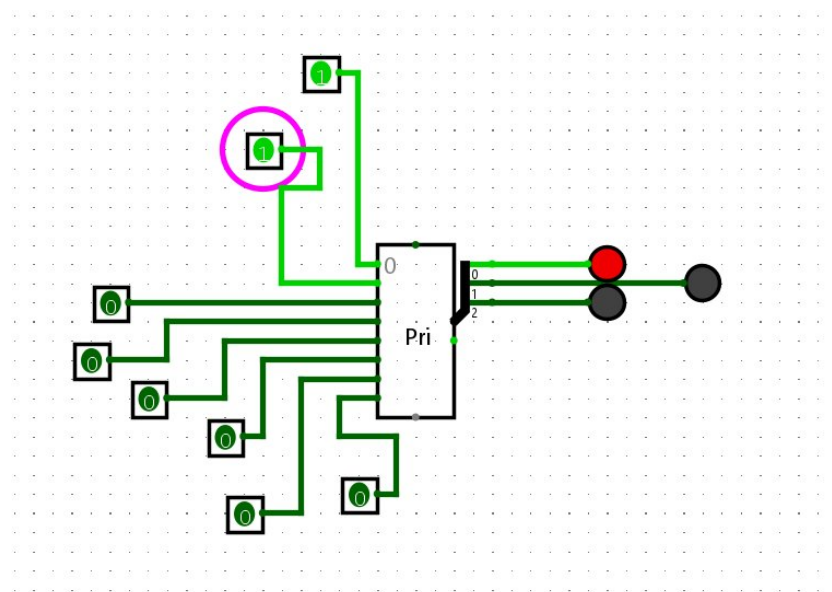


Fig. 4 – Priority Encoder (8×3) using Logisim's built-in priority encoder component.

7. Observations

The 8×1 MUX correctly routed the selected input to the output for all 8 combinations of select lines, matching the expected truth table. The 8×3 encoder produced the correct 3-bit binary output for each active-HIGH input. The priority encoder was also verified using Logisim's built-in component, producing correct priority-based outputs.

8. Result

The 8×1 Multiplexer and 8×3 Encoder were successfully designed and verified on Logisim using basic logic gates. All outputs matched the theoretical truth tables, confirming the correctness of the circuit designs.